Algebra 3

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Contents

1 Rings

1.1 Monoid rings

Definition 1.1.1. A **ring** is an abelian group $(R, +, 0)$ which is also a monoid $(R, \cdot, 1)$ such that $\cdot$ distributes over $+$. We may also require $0 \neq 1$.

Some classical examples are:

- $\mathbb{Z}$, the ring of integers;
- any field like $\mathbb{F}_{p^n}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$;
- $M_n(R)$ the ring of $n \times n$ matrices with entries in R, the first noncommutative example here;
- R-valued functions out of any set have a pointwise ring structure;
- the first Weyl algebra is roughly "the ring of polynomial valued differential operators", defined as

$$A_1 := \mathbb{C}[x, \partial]/(\partial x - x\partial = 1)$$

you should think of this acting of $f \in \mathbb{C}[x]$, generated by the product rule:

$$\partial x f - x\partial f = \frac{d}{dx}(xf) - x\frac{d}{dx}f = f.$$

Definition 1.1.2. For a monoid P and a ring R the **monoid ring** is the set of finite R-linear sums:

$$\sum_{p \in P} r_p p.$$

The addition and multiplication come from the independent inclusions $R, P \subset R[P]$ by $r \mapsto r1_p$ and $p \mapsto 1_R p$ respectively.

Obviously, every group is a monoid, but here are some other examples:

- $\mathbb{N}$ is a semiring, so a monoid under addition;
- given a set X, both $\cup$ and $\cap$ make $\mathcal{P}(X)$ a monoid with $\emptyset, X$ the respective identities;
- the set of endomorphisms of an object in a category is always a monoid;
- let $C \subseteq \mathbb{R}^n$ be a convex cone, i.e. $\mathbb{R}_{\geq 0} C = C$ and $C + C = C$, then $P = C \cap \mathbb{Z}^n$ is a monoid under addition, if $C = \langle (1,0), (-1,2) \rangle$ is generated by poitns in $\mathbb{Z}$ we may not necessarily have P generated as a monoid by these same elements (in this case $(0,1) \in P$ but cannot be realised as a $\mathbb{Z}$-linear combination of $(1,0)$ and $(-1,2)$).

Most of the canonical examples of rings missed out earlier was because they can be realised as monoid rings:

- $R[x_1, \dots, x_n]$ the ring of polynomials in n variables, is just the monoid ring $R[\mathbb{N}^n]$;
- $R[x_1, x_1^{-1}, \dots, x_n, x_n^{-1}]$ the ring of Laurent polynomials in n varialbes, is the monoid ring $R[\mathbb{Z}^n]$;
- we will see that a lot of ring homomorphisms are induced by monoid homomorphisms they are rings over, a first example of this is the isomorphism $\mathbb{C}[\mathbb{Z}/n\mathbb{Z}] \cong \mathbb{C}[x]/(x^n - 1)$;
- we can define the quaternions $\mathbb{H}$ as a quotient of $\mathbb{R}[Q_8]$, where Q_8 is the quaternionic group $\{\pm 1, \pm i, \pm j, \pm k\}$, by the ideal $(1 + (-1), i + (-i), j + (-j), k + (-k))$;
- $\mathbb{R}[P]$ from earlier is a subring of $\mathbb{R}[x, x^{-1}, y, y^{-1}]$ which, as P isn't generated by $\{(1,0), (-1,2)\}$, is actually $\mathbb{R}[x, y, y^2/x]$.

Definition 1.1.3. For two rings S, R a **ring homomorphism** is a function $f : R \to S$ such that f is a homomorphism of the additive groups and multiplicative monoids.

Constructing ring homomorphisms is in general very hard, it is a lot of data.

Definition 1.1.4. If R is a ring and $f : P \to Q$ is a monoid homomorphism then there is an **induced homomorphism** $f_* : R[P] \to R[Q]$ given by:

$$\sum_{p \in P} r_p p \mapsto \sum_{p \in P} r_p f(p).$$

This makes a lot of interesting ring homomorphisms, anything more interesting belongs to the land of algebraic geometry, we will not discuss that here.

The **image** and **kernel** of a ring homomorphism are inhereted from the additive group homomorphism.

Definition 1.1.5. An additive subgroup $I \leq R$ is a **left ideal** if $RI \subseteq I$. Right ideals and two-sided ideals are defined obviously.

The kernel of a ring homomrophism is certainly a two-sided ideal. Conversely, to any ideal I there is a unique ring structure on R/I that makes $\varphi : R \to R/I$ a ring homomorphism.

Definition 1.1.6. A **left unit** in R is an element $u \in R$ such that there exists some $v \in R$ with $uv = 1$.

Are left units always right units? We know this to be true in commutative rings and $M_{n \times n}(k)$, but if we consider $GL(\mathbb{R}^{\mathbb{N}})$, with the basis $\{e_1, e_2, \dots\}$, then the linear map which sends each e_i to e_{i+1} is injective and has a left inverse, but is not surjective so has no right inverse. But, if u has both a right inverse **and** a left inverse, then these will always be the same. The group of two-sided units is called $R^\times$.

1.2 Classical algebra and whatnot

You probably already know what an Euclidean domain, principal ideal domain, unique factorisation domain, and a Noetherian ring are.

There won't be a particularly rigorous discussion of the relationship, because I don't care. Look at any set of official notes for something of higher quality if you require it.

Proposition 1.2.1. An Euclidean domain is a principal ideal domain.

Proof. Let R be an Euclidean domain with degree function φ, and let $I \trianglelefteq R$, consider $\varphi(I)$, this must have a smallest element, call it a. Now for any $i \in I$, we can write $i = qa + r$ with $\phi(r) < \phi(a)$, but r must be in I so cannot have degree less than that of a, so $r = 0$. Therefore, $I = (a)$. □

Proposition 1.2.2. A ring is Noetherian iff all ideals are finitely generated.

Proof. Suppose an ideal $I \trianglelefteq R$ does not admit a finite generating set. Find $i_0 \in I$, and $i_1 \in I \setminus (i_0)$, and recursively find $i_n \in I \setminus (i_0, i_1, \ldots, i_{n-1})$, this produces an ascending chain:

$$(i_0) \subsetneq (i_0, i_1) \subsetneq (i_0, i_1, i_2) \subsetneq \cdots$$

Conversely, suppose every ideal in R is finitely generated, then for any ascending chain of ideals:

$$I_0 \subset I_1 \subset I_2 \subset \cdots$$

their union I is also an ideal, which is finitely generated with its elements living in some I_N, thus the chain stabilises at this ideal. □

Proposition 1.2.3. A principal ideal domain is a unique factorisation domain.

Proof. First we show that any $r \in R$ a PID can't be an infinite product of irreducibles. Suppose such and r exists, then as r irreducible we can find a nontrivial factorisation $r = r_1 s$, one of these must contain infinitely many irreducibles, let this be r_1. Repeat this process ad infinum to get a chain of ideals

$$(r) \subsetneq (r_1) \subsetneq (r_2) \subsetneq \cdots$$

as R is a PID it is also Noetherian, so this ascending never stabilising chain of ideals cannot exist.[1]

We now wish to show any such product of irreducibles

$$r_1 r_2 \cdots r_n = r = s_1 s_2 \cdots s_m$$

are equivalent. First note that in a PID an element is irreducible iff it is prime.[2] We can now see each r_i appears in the RHS, as we are living in an integral domain we can cancel and thus are done. □

1.3 Unique factorisation in polynomial rings

Definition 1.3.1. Let R be a UFD, if $f \in R[x]$ the **content** of f is the gcd of its nonzero coefficients, written $c(f)$.

Lemma 1.3.2. If $f, g \in R[x]$, then $c(fg) = c(f)c(g)$.[3]

This really follows easily from

Lemma 1.3.3. If $c(f) = c(g) = 1$ then $c(fg) = 1$.

Proof. Suppose $c(fg) \neq 1$, then there exists some prime p dividing all the coefficients. First observe that as R is an integral domain, so is $R[x]$. Now consider the projection mod p, as $c(f) = c(g) = 1$ there exists no prime dividing all elements so they are both nonzero in $R/(p)[X]$ which we know to be an ID, but supposedly their product is 0. This is clearly a contradiction. □

[1] This proof is clearly nonconstructive as we are choosing contably many irreducible factorisations all at once

[2] Given r irreducible, if $r \mid ab$ we can consider $c = \gcd(r, a)$ and observe $c \mid r$. I'm only really interested in R an ED, so lets assume that. We know $r = cd$ for some d, as r is irreducible either c is a unit, in which case we can write $xr + ya = c$ implying $xrb + yab = cb$, as $r = ab$ we have that $r \mid b$. If, instead, c is an associate of r, then $r \mid a$. The other direction is easier.

[3] As with lots of things in this course, this must be taken up to multiplication by a unit.

For our next step we need the following

Lemma 1.3.4 (Gauss lemma)**.** If R is a UFD, and $f \in R[x]$ splits in $\mathrm{Frac}(R)[x]$, then f splits in $R[x]$.

Proof. If $f = gh$ with $g, h \in \mathrm{Frac}(R)[x]$, then we can find $a, b \in R$ with $ag, bh \in R[x]$ with $c(ab) = c(bh) = 1$, so $c(abf) = 1$ even though all coefficients in f are multiples of ab, so ab is a unit and thus $g, h \in R[x]$. □

Theorem 1.3.5. If R is a UFD then so is $R[x]$.

Proof. For the existence of a factorisation, we can write $f = c(f)f'$ where f' is primitive, now we can factorise $c(f)$ as in R and now all factorisations of f' strictly decrease the degree, and the number of factors is bounded by the degree, so we will get a finite factorisation into irreducibles.

For uniqueness, likewise we already know the factorisation of the content is unique. Now by the Gauss lemma we know the factorisation of f' into irreducibles in $R[x]$ is the same as that in $\mathrm{Frac}(R)[x]$. So if p and q are irreducible components in two possible factorisations of f' which are similar in $\mathrm{Frac}(R)[x]$ we can find $a, b \in R$ with $ap = bq$ so $ab \mid c(ap)$ which implies $b \mid c(p)$ so b, and thus a are both unit, and our factorisation is unique up to units. □

1.4 Newton polytopes

If I have a polynomial $f \in \mathbb{C}[x_1, \dots, x_n]$, then this corresponds to a set of points in $\mathbb{N}^n$ labelled with the monomials coefficient in $\mathbb{C}$. Call the convex hull of these points the **Newton polytope** of f. If $f = gh$ then the Newton polytope of f is the Minkowski sym of the Newton polytope of g and h, this is completely obvious from the definition of the moinoid ring. Drawing these convex hulls and determining if they are decomposable and then solving a system of linear equations is a robust method for decomposing multivariate polynomials into irreducibles.

1.5 Hilbert basis theorem

2 Invariant theory

All rings are commutative.

Definition 2.0.1. If K is a field, a **k-algebra** is a ring R continaing k. If $k \subset R_1, R_2$ are two such k-algebras, then a **k-algebra homomorphism** is a ring homomorphism which is the identity when restircted to k.

$R = \mathbb{C}[X_1, \dots, X_n]$ is a $\mathbb{C}$ algeabra, $\mathbb{Z}$ is not a k-algebra for any k.

Definition 2.0.2. A k-algebra R is **finitely generated** if there exists a finite subset $S\{a_1, \dots, a_n\} \subset R$ such R is the smallest k-subalgebra containing S.

Equivalently, if there exists a k-algebra homomorphism

$$\mathrm{ev}_S : k[X_1, \dots, X_n] \to R$$

which send f to $f(a_1, \dots, a_n)$ and is surjective.

So all f.g. k-algebras are rings of the form

$$k[X_1, \dots, X_n]/I$$

for some ideal I which, by the Hilbert basis theorem, will be finitely generated. We like these, because *in principle* every questions about f.g. k-algebra (of which there are many of great intereset) can be answered algorithmically.

Even thought there are many interesting k-algebras which aren't finitely generated, like $\mathbb{C}\{X\}$ the ring of power series in complex coefficients with radius of convergence $R > 0$, so we can do pointwise operations.

The group $GL_n(\mathbb{C})$ has a **right**[4] action on $\mathbb{C}[X_1, \dots, X_n]$ by acting on $\mathbb{C}^n$ as matrices before applying the polynomial.

[4] People really don't like right actions, they like turning them into left actions by consider G^{op} or explicitly acting by inverses, we want to be able to efficiently compute matrices involving our actions, so we will just tolerate working with right actions.

Definition 2.0.3. A **graded k-algebra** is a k-algebra

$$R = R_0 \oplus R_1 \oplus \cdots = \bigoplus_{n \in \mathbb{N}} R_n$$

such that $R_0 = k$ and for each $n \geq 1$, R_n is a finite dimension k-vector space, and for all $n, m \in \mathbb{N}$, $R_n R_m \subset R_{n+m}$.

A **graded k-algebra homomorphism** $f : R \to S$ should satisfy $f(R_n) \subseteq S_n$ for all $n \in \mathbb{N}$.

$k[X_1, \ldots, X_n]$ is the obvious example of a graded k-algebra by the degree of the polynomials, and the action of $GL_n(k)$ is in fact graded, as it preserves polynomial degree.

In general, graded k-algebras don't satisfy a first isomorphism theorem, the cokernel and image can disagree on what degree something's in.

2.1 Finite generation

Proposition 2.1.1. For $G \leq GL_n(\mathbb{C})$ a finite subgroup acting on $R = \mathbb{C}[x_1, \ldots, x_n]$, the set of G-invariant polynomials R^G is a ring.

Proof. Follow your nose. □

Definition 2.1.2. We have the **Reynold's operator** $\rho : R \to R^G$ given by:

$$\rho(f) = \frac{1}{|G|} \sum_{g \in G} f^g$$

This is basically an average over G. It is **not** a ring homomorphism

Lemma 2.1.3. ρ is a graded R^G-module homomorphism, and a projection onto R^G.

Theorem 2.1.4. Under the same assumptions, R^G is in fact finitely-generated as a $\mathbb{C}$-algebra.

Proof. Consider the ideal I generated by $R^G \setminus k$, by Hilbert's basis theorem, I is finitely generated by some $f_1, \ldots, f_m \in I$. Each f_i must be of the form

$$f_i = \sum g_{ij} h_{ij}$$

with each $g_{ij} \in R$ and $h_{ij} \in R^G \setminus k$. By decomposing these into their homogeneous terms, I may assume also assume all the generators are homogeneous, call this set $r_1, \ldots, r_l$ where each $r_i \in R^G_{m_i}$.

Now, if I let S be the k-subalgebra generated by $r_1, \ldots, r_l$, I know $S \subset R^G$, and now I'll prove by induction on N that $R^G_N \subset S$, the base case is just $k \subset S$.

Pick $f \in R^G_N$, then $f \in I$ and so I can write

$$f = \sum_{i=1}^{l} f_i r_i$$

with each of the $f_i \in R$ homogeneous of degree $N - m_i$ (as f and r_i are homogeneous, anything else will get cancelled away, so I may as well not include it). Now I can use the Reynold's operator to write

$$f = \rho(f) = \sum \rho(f_i) r_i$$

where, as ρ preserves the grading, $\rho(f_i) \in R^G_{N-m_i}$ so, by induction, is in S so generated by $r_1, \ldots, r_s$, hence f is generated by the $r_1, \ldots, r_l$ thus in S. □

This proof is very concise, and very clean. But it is basically useless, as it is not *effective*! We have no way to actually produce the finite set of generators for R^G, just the incorporeal promise of their existence somewhere. We should find a **better** proof.

Theorem 2.1.5. R^G is generated by the set of all elementary symmetric polynomials on $\{x_i^g\}_{g \in G}$ for each index i, and $\rho(x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n})$ for all $0 \leq \alpha_1, \ldots, \alpha_n \leq |G|$.

Proof. First we show that R is genereated by all the elementary symmetric polynomials on $\{x_i^g\}_{g\in G}$ and the same monomials $x_1^{\alpha_1}x_2^{\alpha_2}\cdots x_n^{\alpha_n}$ with highest power $\leq |G|$, without applying the Reynold's operator.

For each x_i, consider the polynomial:

$$P_i(T) = \prod_{g\in G}(T - x_i^g) = T^{|G|} + a_{i,1}T^{|G|-1} + \cdots + a_{i,|G|}$$

the coefficients $a_{i,j}$ are exactly all the elementary symmetric polynomials on $\{x_i^g\}_{g\in G}$. For each i, $P_i(x_i) = 0$, thus:

$$x_i^{|G|} = -a_{i,1}x_i^{|G|-1} - \cdots - a_{i,|G|}$$

Hence, by induction, we can always strictly decrease the degree of any polynomial $f \in R$ until it is just the sum of monomials $x_1^{\alpha_1}x_2^{\alpha_2}\cdots x_n^{\alpha_n}$ with highest power $\leq |G|$.

Now, for any $f \in R^G$, we write:

$$f = \rho(f) = \rho\left(\sum a_{ij}x_1^{\alpha_1}\cdots x_n^{\alpha_n}\right) = \sum a_{ij}\rho(x_1^{\alpha_1}\cdots x_n^{\alpha_n})$$ □

2.2 Elementary symmetric polynomials

We know the elementary symmetric polynomials e_i in n variables

$$\prod_{i=1}^{n}(T - x_i) = \sum_{i=0}^{n}(-1)^i T^{n-i} e_i(x_1, \dots, x_n) \in \mathbb{Z}[x_1, \dots, x_n][T]$$

which can write a bit more explicitly as

$$e_0 = 1, \qquad e_1 = \sum_{i=1}^{n} x_i, \qquad e_2 = \sum_{1\leq i<j\leq n} x_i x_j, \qquad e_k = \sum_{1\leq i_1<\cdots<i_k\leq n} x_1\cdots x_l \quad \text{for } k \leq n$$

with $e_k \in \mathbb{Z}[x_1, \dots, x_n]$ homogeneous of degree k.

The symmetric group S_n has an induced action on $\mathbb{Z}[x_1, \dots, x_n]$ in the same way $GL_n(\mathbb{C})$ does on $\mathbb{C}[x_1, \dots, x_n]$, and all the $e_i(x_1, \dots, x_n) \in \mathbb{Z}[x_1, \dots, x_n]^{S_n}$.

Theorem 2.2.1. Consider $\mathbb{Z}[y_1, \dots, y_n]$ as a graded ring where $\deg(y_k) = k$, then the homomorphism $\phi : \mathbb{Z}[y_1, \dots, y_n] \to \mathbb{Z}[x_1, \dots, x_n]^{S_n}$ given by $\phi(y_k) = e_k(x_1, \dots, x_n)$ is an isomorphism of graded rings i.e. the elementary symmetric polynomials generate $\mathbb{Z}[x_1, \dots, x_n]^{S_n}$ and there are no algebraic relations between them.

Proof. Order the monomial in $R = \mathbb{Z}[x_1, \dots, x_n]$ lexicographically (I care about the powers of x_i more than x_j when $i < j$). For all $f \in R$, I have a highest monomial $hm(f) = x^m$, if $f \in R^{S_n}$, then $m = (m_1, \dots, m_n)$ must satisfy $m_1 \geq m_2 \geq \cdots \geq m_n$. I now have two remarks: $hm(e_k) = x_1x_2\cdots x_k$, and $hm(fg) = hm(f)hm(g)$, from which we can conclude

$$hm(e_1^{k_1}e_2^{k_2}\cdots e_n^{k_n}) = (x_1^{k_1})(x_1^{k_2}x_2^{k_2})\cdots = x_1^{k_1+k_2+\cdots+k_n}x_2^{k_2+\cdots+k_n}\cdots x_k^{x_k}$$

from this, we can convince ourselves that every highest monomial of a symmetric polynomial is a scalar multiple of the highest monomial of something in $\operatorname{im}\phi$.

We can now use a slightly involved induction procedure, let f have degree d, then there are finitely many monomials with degree d which are ordered lexicographically, if we subtract the product of elementary symmetric polynomials with the same leading coefficient as f, we will strictly decrease the order of $hm(f)$ among monomials of degree d, or strictly decrease the degree of f, so it requires a finite number of steps to strictly decrease the degree of f, now do induction on d.

To show ϕ is injective we just note that the leading monomial of two products of elementary symmetric polynomials are equal iff the two products are equal, so they can never have algebraic relations. □

2.3 Groebner bases

Throughout this and the next secion, use the lexicographic ordering on monomials in polynomial rings.

If we modify our statement of the division algorithm in $K[x]$ to be that for all $f, g \in K[x]$ with $g \neq 0$, there is a unique expression $f = qg + r$ where no term of r is divisible by the **leading term** $LT(f)$, of f. This way of phrasing things generalises nicely to the multivariate case.

If we want to divide $f \in K[x_1, \dots, x_n]$ by $f_1, \dots, f_s \in K[x_1, \dots, x_n]$, then we want to find an expression $f = q_1 f_1 + \cdots + q_s f_s + r$ where no term of r is divisible by any of the $LT(f_1), \dots, LF(f_s)$. In order to minimise cancellation we should also expect $LT(f) \geq LF(q_i f_i)$ for all i, such an expression is called **standard**.

For a general choice of $f_1, \dots, f_s$, the remainder is not unique up to redordering of the f_i, consider $xy^2 - x$ divided by $xy + 1$ and $y^2 - 1$.

Definition 2.3.1. If $I \trianglelefteq K[x_1, \dots, x_n]$, we say $\{g_1, \dots, g_s\} \subseteq I$ is a **Groebner basis** for I if

$$(LT(g_1), \dots, LT(g_s)) = (\{LT(f) \mid f \in I\})$$

as ideals in $K[x_1, \dots, x_n]$.

These ideals generated by monomials have good combinatorial properties, they are called **monomial ideals**.

Proposition 2.3.2 (Dickson's lemma)**.** All monomial ideals have finite bases consisting of monomials.

First Proof. Hilbert's basis theorem gives a finite basis, we can write each element as a finite sum of monomials in the generating set, these monomials for a finite basis for the whole ideal. □

Second Proof. Do induction on n, for the base case we have a monomial ideal I in $K[x]$, we can just take the least n such that $x^n \in I$.

Now work in the ring $S = K[x_1, \dots, x_{n-1}, y]$, given a monomial ideal $I \trianglelefteq S$, consider the ideal J generated by the set $\{x^\alpha \in K[x_1, \dots, x_{n-1}] \mid x^\alpha y^i \in I \text{ for some } b \in \mathbb{N}\}$, by the inductive hypothesis we find a finite generating set $J = (x^{\alpha_1}, \dots, x^{\alpha_s})$ of monomials, where for each i there exists some $b_i \in \mathbb{N}$ with $x^{\alpha_i} y^{b_i} \in I$, define $m := \max\{m_1, \dots, m_s\}$, we can now subdivide J as

$$J = \bigcup_{i=0}^{m} J_i \qquad J_i := (\{x^\alpha \in K[x_1, \dots, x_{n-1}] \mid x^\alpha y^i \in I\}) = (x^{\alpha_i(1)}, x^{\alpha_i(2)}, \dots, x^{\alpha_i(s_i)})$$

where for each i, the $x^{\alpha_i(1)}, x^{\alpha_i(2)}, \dots, x^{\alpha_i(s_i)}$ are a finite generating set of monomials, I claim these $x^{\alpha_i(j)}$ generate I. For all monomials $x^u y^j \in I$, if $j \geq m$, then $x^u \in J$, so $x^{\alpha_m(i)} y^m \mid x^u y^j$ for some i, if instead $j < m$, then $x^u \in J_j$ and hence there exists some i with $x^{\alpha_j(i)} y^j \mid x^u y^j$. □

If I is generated by $\{x^\alpha \mid \alpha \in A\}$ for some $A \subseteq \mathbb{N}^n$, then we may assume any finite generating set is also 'contained' in A, this will be the same argument as for homogeneous generators in the first proof of finite generation of the ring of invariants.

Theorem 2.3.3. For any ideal $I \subseteq K[x_1, \dots, x_n]$, there exists a Groebner basis for I, all Groebner bases of I are also regular bases of I.

First Proof. We know $(LT(I)) := (\{LT(f) \mid f \in I\})$ is a monomial ideal, so it admits a finite monomial basis $h_1, \dots, h_s$, each h_i can be taken to be $LT(g_i)$ for some $g_i \in I$, thus forming a Groebner basis.

For all $f \in I$, write $f = q_1 g_1 + \cdots q_s g_s + r$ where no term of r is divisible by an $LT(g_i)$, if $r \neq 0$, then $LT(r) \in (LT(I)) = (LT(g_1), \dots, LT(g_s))$, and so by our earlier discussion, $LT(r)$ is divisible by one of the $LT(g_i)$ a contradiction, hence r is 0. □

If we come prepared with a finite generating set $I = (f_1, \dots, f_r)$ then we have a nice algorithm for computing the Groebner basis, called **Buchberger's algorithm**.

Definition 2.3.4. For $f, g \in K[x_1, \dots, x_n]$ the S**-polynomial** is

$$S(f_1, f_2) = \frac{\operatorname{lcm}(LT(f), LT(g))}{LT(f)} f - \frac{\operatorname{lcm}(LT(f), LT(g))}{LT(g)} g$$

Theorem 2.3.5 (Buchberger's criterion)**.** A basis $G = \{g_1, \dots, g_s\} \subseteq I$ is a Groebner basis for I iff for all $i < j$, if we divide $S(i, j)$ by G, we get 0.

Now given our finite generating set, we can compute all the S-polynomials, and keep adding the remainders when divided by G.

Both the proof of Buchberger's criterion, and the fact Buchberger's algorithm terminate are non-examinable.

2.4 Elimination theory

Theorem 2.4.1. If $I = (f_1, \dots, f_r) \trianglelefteq K[x_1, \dots, x_n]$ and $G = \{g_1, \dots, g_s\}$ is a Groebner basis for I, then for all $2 < k < n$, $G \cap K[x_k, \dots, x_n]$ is a Groebner basis for the ideal $I \cap K[x_k, \dots, x_n]$.

Proof. For all $0 \neq f \in I \cap K[x_k, \dots, x_n]$, we have $f \in I$, so $LT(g_i) \mid LT(f)$ for some $g_i \in G$, thus $LT(g_i) \in K[x_k, \dots, x_n]$, as we are using the lexicographic order, this mean $g_i \in K[x_k, \dots, x_n]$, thus $G \cap K[x_k, \dots, x_n]$ is a Groebner basis. □

Now if I have computed the ring of invariant R^G for a finite $G \leq GL_n(\mathbb{C})$ acting on $R = \mathbb{C}[x_1, \dots, x_n]$, and have found a finite set of generators $\{f_1, \dots, f_r\}$, then consider the ideal generated by the $f_i - y_i$ in $\mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_r]$, then I can use the previous theorem to compute a generating set of the ideal of relations between my f_i, call this J, and I have $R^G \cong \mathbb{C}[y_1, \dots, y_r]/J$.

3 Modules

3.1 Free modules

3.2 Modules over a PID

4 Matrix Lie groups

Formally, Lie groups are the group objects in the category of smooth manifolds. Many interesting examples and much deep theory of Lie groups is accessable by only considering matrix Lie groups, which avoid developing the full theory of smooth manifolds.

Definition 4.0.1. For a commutative ring R, we write:

$$GL_n(R) := \{n \times n \text{ invertible matrices with coefficients in } R\}$$

and we will almost always only be interested in $GL_n(\mathbb{R})$ and $GL_n(\mathbb{C})$.

These are both topological spaces by considering the inclusion:

$$GL_n(\mathbb{R}) \subset M_{n\times n}(\mathbb{R}) = \mathbb{R}^{n^2}$$

$GL_n(\mathbb{R})$ is in fact an open subset, as the determinant:

$$\det : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$$

is a polynomial in its coefficients, so must be continuous. We can thus write

$$GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R}^\times)$$

so $GL_n(\mathbb{R})$ inehrets a locally Euclidean topology from $\mathbb{R}^{n^2}$.[5]

Definition 4.0.2. A **matrix Lie group** is a closed subgroup of $GL_n(\mathbb{R})$.

Some classical examples of matrix Lie groups include:

$$(\mathbb{R}, +) \cong \left\{\begin{pmatrix} 1 & \lambda \\ 0 & 1\end{pmatrix} \in GL_n(\mathbb{R})\right\} = a^{-1}(1) \cap c^{-1}(0) \cap d^{-1}(1)$$
$$O_n(\mathbb{R}) = \{A \mid A^T A = I\} \subset GL_n(\mathbb{R}) \qquad (\det(O_n) = \pm 1)$$
$$SL_n(\mathbb{R}) = \det^{-1}(1) \subset GL_n(\mathbb{R})$$
$$U_n(\mathbb{C}) = \{A \in GL_n(\mathbb{C}) \mid A^\dagger A = I\} \qquad (\det(U_n) = S_1)$$
$$\mathrm{Eucl}_n^+(\mathbb{R}) = \mathbb{R}^n \rtimes SO_n(\mathbb{R})$$
$$\mathrm{Aff}_n(\mathbb{R}) = \mathbb{R}^n \rtimes GL_n(\mathbb{R})$$
$$\text{finite reflection groups like } D_{2n} \leq O_2$$
$$\text{any finite group } G \leq S_n \leq SL_n$$
$$O(p, q) = \{A \in GL_{p+q}(\mathbb{R}) \mid \text{such that } A \text{ fixes all quadratic forms with inertia } (p, q)\}$$

4.1 Matrix exponential

Definition 4.1.1. If $A \in M_{n\times n}(\mathbb{R})$ define the matrix exponential:

$$\exp(A) = \sum_{i=0}^{\infty} \frac{1}{k!} A^k$$

Proposition 4.1.2. For all matrices $A \in M_{n\times n}(\mathbb{R})$ this converges

Sketch. As all norms on $\mathbb{R}^{n^2}$ are equivalent, we can show convergence by any of them, we'll use the d_∞ norm. Going by absolute convergence

$$\left\|\sum_{i=0}^{n} \frac{1}{k!} A^k\right\| \leq \sum_{i=0}^{n} \frac{1}{k!} \left\|A^k\right\| \to e^{\|A\|}$$

gives us our bound, we then show the sequences is Cauchy. □

Theorem 4.1.3. We can in fact do even better. Not only is exp continuous, it is continuously differentialable, and all derivates converge uniformly on all compact subsets of the domain.

Some obvious first properties of the matrix exponential

[5] The same argument applies for $GL_n(\mathbb{C})$, I often won't include this in the future.

4.2 Lie algebras

Theorem 4.2.1. $f : G_1 \to G_2$ a continuous group homomorphism of matrix Lie groups induces a homomorphism of Lie algebras $f_* : \mathfrak{g}_1 \to \mathfrak{g}_2$.